

Qualitative Comparison: Success & Failure Cases

(a) Success: Factual Generation

Prompt: "The discovery of gravitational waves in 2015"

DFlow-LM: "...by the LIGO collaboration confirmed a century-old prediction of Einstein's general theory of relativity. The detection of ripples in spacetime opened a new era..."

MDLM: "...was the the important waves of the detection physics physics important the the waves the the important..."

(b) Success: Text Infilling

Context: "The cat sat on the _____ and watched."

DFlow-LM: "warm windowsill"
(natural, contextually appropriate)

MDLM: "the the sat"
(degenerate, repetitive)

(c) Failure: Long-Range Repetition

After 400+ tokens of coherent generation:

DFlow-LM: "...the city was known for its the city of the city of the city..."

Occurs in <5% of long (>512 tok) generations.
Mitigated with repetition penalty.

(d) Failure: Code Generation (OOD)

Prompt: "def fibonacci(n):"

DFlow-LM: "def fibonacci(n): the the function of the the return..." (invalid)

Expected: single-domain (Wiki) model
cannot generate valid code.